College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

**Midterm Examination of Semester 2** **Year**: 2017

**Subject**: 340152 Electrical Measuring Instruments **Section**: 5-6

**Date**: 9 March 2018 **Time**: 8.00-10.00

Name_____ID: _____Class_____

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**Instructions:**

1. The examination has 8 pages (including this page) and a total score of 70 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a **closed book** examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

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**Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.**

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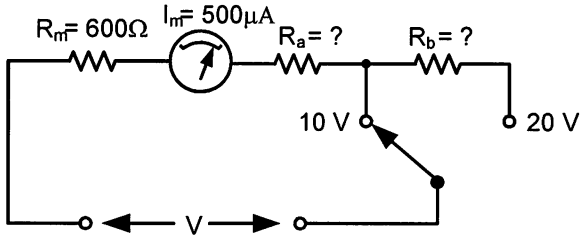
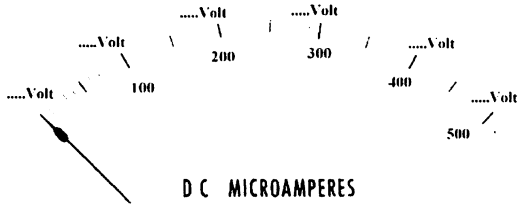
**Q1)** A PMMC instrument with FSD = 500 μA and $R_m$ = 600Ω, is to be employed as a volt-meter with ranges of 10 V, and 20 V. (10 marks)

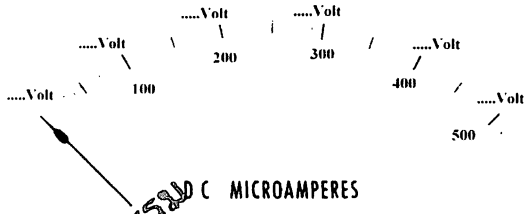
(1.1) Calculate the required values of multiplier resistors for the circuit.




(1.2) Determine how the voltmeter scale should be marked.




(a) 10 V




(b) 20 V

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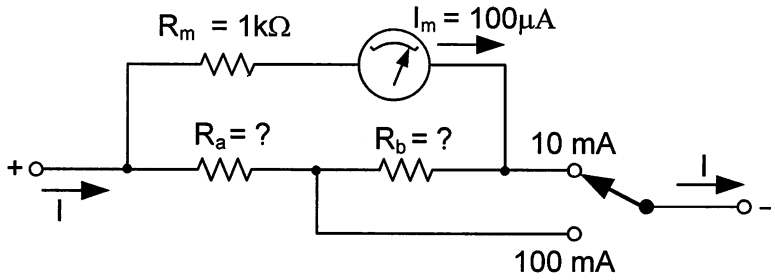
**Q2)** A PMMC instrument has 100 μA and a coil resistance of 1kΩ. Calculate the required resistance values to convert the instrument into **an ammeter (multi-range)** with an FSD of (a) 10 mA (b) 100 mA. (10 marks)

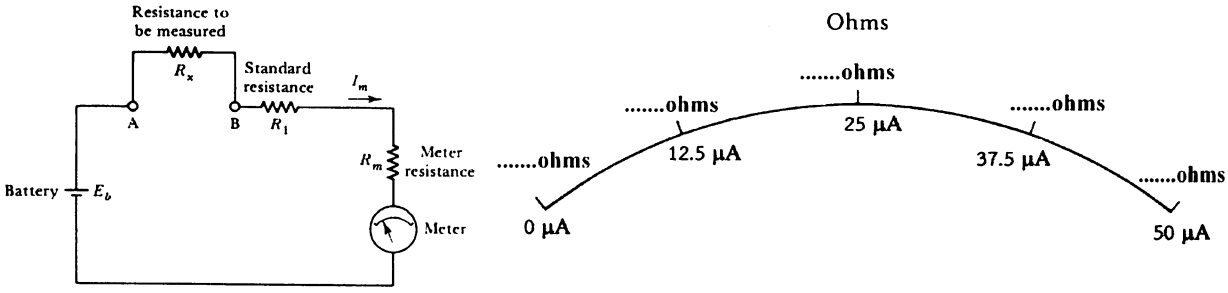


**Q3)** The series ohmmeter in figure is made up of a 1.5 V battery, a 50 μA meter, $R_m$ = 1.2 kΩ.

(a) Calculate the value of $R_1$ to make the current to be full scale.

(b) Determine how the resistance scale should be marked at $R_x = 0$ and $R_x = \infty$.

(c) Determine how the resistance scale should be marked at 0.75 FSD, 0.5 FSD, and 0.25 FSD. (10 marks)

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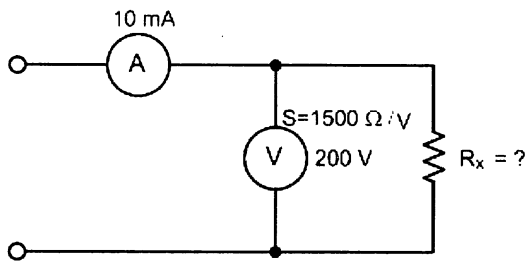
**Q4)** A voltmeter having a sensitivity of 1,500 Ω/V, reads 200 V on its 300 V scale when connected across an unknown resistor (Rx) in series with an ammeter and the ammeter reads 10 mA, Calculate the following items:

(a) The apparent resistance of the unknown resistor ($R_T$),

(b) The actual resistance of the unknown resistor ($R_X$),

(c) % error due to the loading effect. (10 marks)

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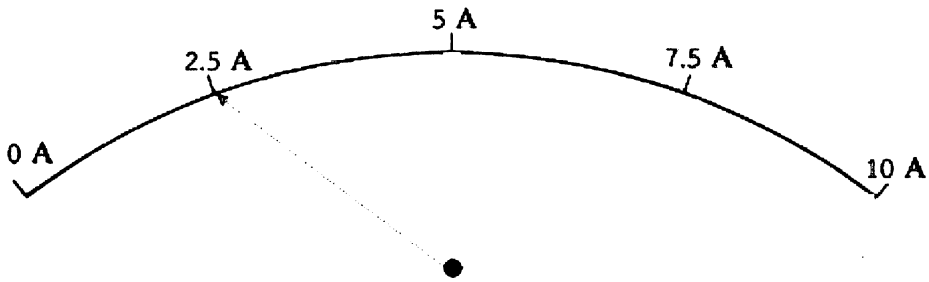
**Q5)** An Ammeter **range 10A class 2**, Calculate the maximum %Error for position of pointer **2.5 A.** (4 marks)

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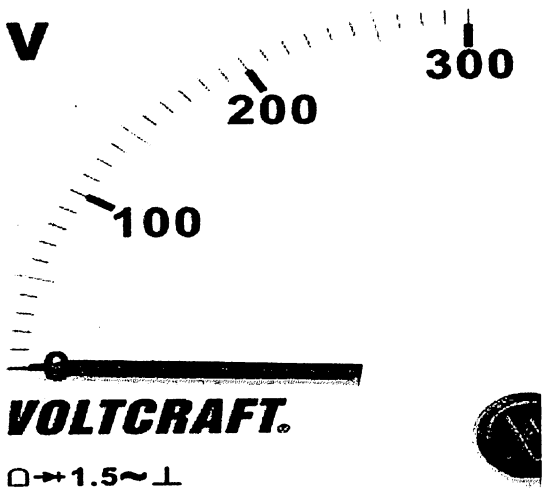
**Q6)** Identify each of the symbols illustrated in Figure. (3 marks)




a) ⏛ : .....

b) 1.5 : .....

c) ⊥ : .....

**Q7)** Define the following terms: (5 marks)

(a) Error :.....

(b) Accuracy :.....

(c) Precision :.....

(d) Measurement :.....

(e) Instrument : .....

**Q8)** State the three major categories of error. (2 marks)

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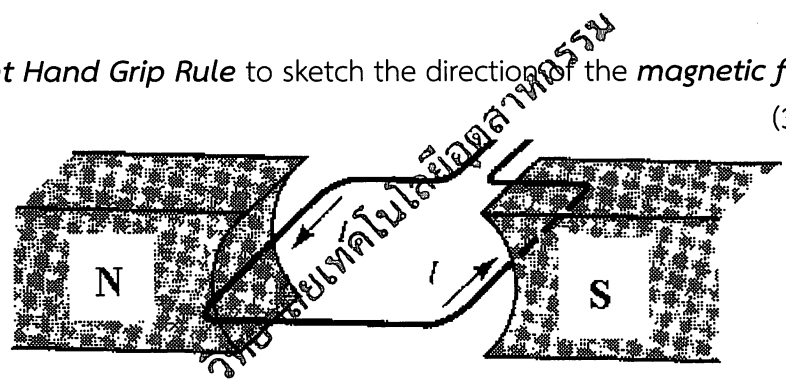
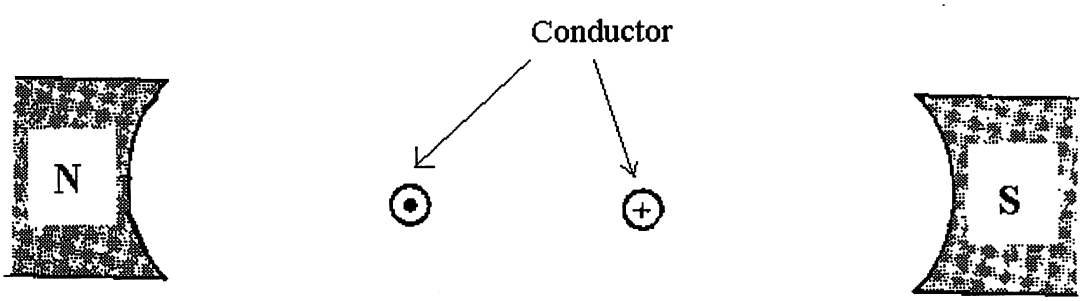
**Q9)** Using ***Right Hand Grip Rule*** to sketch the directions of the ***magnetic field*** and ***force***. (3 marks)




**Q10)** List and explain the **three forces** involving in the moving system of a deflection instrument. (3 marks)

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**Q11)** Describe the basic concept of meter movement. (3 marks)

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**Q12)** The expected value of a resistor is 4.7 kΩ, but the measurement using ohmmeter yields a value of 4.65 kΩ. Calculate (a) absolute error, (b) % error, (c) accuracy, and (d) % of accuracy. (5 marks)

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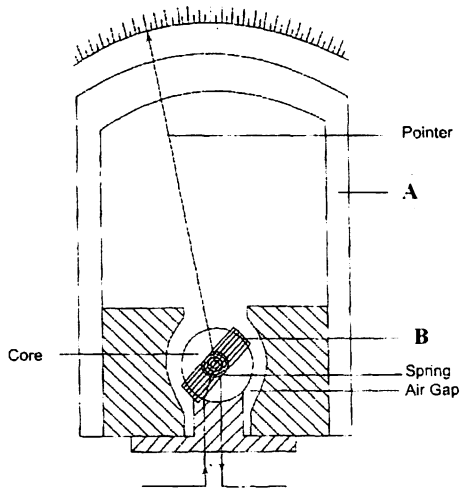
**Q13)** Identify each part of the instrument: (2 marks)




Part A is .....

Part B is.....

Assoc. Prof. Dr. Pichet Sriyanyong